

Low Pressure Chemical Vapor Deposition (LPCVD) Reactor: COMSOL Simulation

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Abstract

Chemical Vapor Deposition (CVD) involves a chemical reaction of a gaseous reactant adsorbed onto a surface and forms a solid product. This process has been used in thin film production in various different industries. In this experiment, a CVD at low pressure, or LPCVD, is simulated in COMSOL to observe the effect on silicon deposition rate as operating conditions are deviated from optimal circumstances. 2^k factorial design of experiment is used to find the effectiveness factor of the varied parameters. It is concluded that silane mole fraction in the feed stream had greatest impact on average deposition rate, followed by temperature and pressure.

1 Introduction

Chemical vapor deposition (CVD) is a process in which a gaseous reactant stream is fed into the reactor and is adsorbed onto the surface of the reaction site. The result of this reaction is a solid product left on the surface. CVD has been used since approximately the mid-1900s and has undergone improvements in the few decades. Chemical vapor deposition carried out at atmospheric pressure (APCVD) was commonly used until the usage of low pressure was introduced in the 1970s. APCVD is limited by the rate of mass transfer, whereas low pressure chemical vapor deposition (LPCVD) is limited by the rate of reaction. This generally results in increased uniformity, quality, and coverage of observed deposits.¹ One more key advantage of LPCVD is its ability to operate at lower temperatures in which made it more suitable method of deposition after prior layers that may have a low melting point. This operating temperature can also be reduced further with the addition of energy into the reaction, typically a plasma.² Plasma-enhanced chemical vapor deposition (PECVD) is another increasingly popular method and is continuing to be researched. LPCVD can form deposits ranging from nanometers to micrometers and has much versatility in comparison to other methods of thin film deposition. Because of this, LPCVD is commonly utilized to produce thin films in the microelectromechanical systems and semiconductor industry.³ Some other notable applications of LPCVD can be found in coating of tools and diamonds as well as the automobile and aerospace industry.

2 Background

The modeling of single-wafer LPCVD reactors have been carried out as early as 1989. Problems in the single-wafer design which this early model addresses include the lack of uniformity of the deposited film and the change in thermodiffusion driving large silane molecules away from the substrate.⁴ The wafer in the reactor used had a diameter of 0.25 meters and was subject to an impinging jet of silane vapor.⁴ To increase the surface area thereby increasing the deposition rate, it is beneficial to have larger wafers in this design, but an alternative is to use

many smaller wafers spaced apart so that the vapor diffuses from the convective stream to the space between the wafers and deposits onto the substrate. A benefit of the design using many smaller wafers is diffusion being partly driven by the radial diffusion of the vapor in addition to the axial convection. By lining the middle portion of the reactor with a series of wafers, a smaller axial cavity can be made, allowing a greater vapor flow rate through the portion of the reactor where solid deposition takes place.

The temperature of the vapor and its concentration have effects on the deposition rate and the method by which the deposition rate can be evaluated. Wafers in LPCVD reactors contain microtrenches to which the growth species adheres. At lower temperatures, the sticking probability of the growth species decreases, and the overall process may become limited by the surface reaction whereas at higher temperatures, the rate of deposition is limited by reactions in the gas phase. This temperature was determined by Komiyama et al. to be roughly 150°C for the deposition of silicon using silane vapor.⁵ At lower concentrations of the gas-phase, the reaction rate decreases, and the overall process becomes diffusion-limited.⁵ The models used to evaluate the deposition rate should account for the change in the rate-limiting step at certain regimes of vapor concentration and temperature.

In our study, we use 2^k factorial design to determine the effect of vapor properties and reactor design on the silicon deposition rate in COMSOL. The reactor was modeled using separate physics simulations of fluid convection, heat transfer, and vapor diffusion in order to evaluate the relative effects of each parameter.

3 Theory

The equations used in the COMSOL model are simplifications of transport principles. Fluid motion in the reactor is described as

$$\rho(\vec{u} \cdot \nabla)\vec{u} = \nabla \cdot [-p\vec{I} + \vec{K}] + \vec{F} \quad (1)$$

for $\vec{u}$ velocity field, p pressure, fluid density ρ where $\rho \vec{l}$ is a nondimensionalization of internal forces, $\vec{K}$ is the viscous stress tensor, and $\vec{F}$ is the contribution of external forces, where the assumptions that fluid motion is always laminar flow and that the fluid is incompressible

$$\rho \nabla \cdot \vec{u} = 0 \quad (2)$$

were made. Heat transfer is described as

$$\rho C_p \vec{u} \cdot \nabla T + \nabla \cdot \vec{q} = Q + Q_p + Q_{vd} \quad (3)$$

along with Fourier's law

$$\vec{q} = -k \nabla T \quad (4)$$

for C_p heat capacity of the fluid, k fluid conductivity, where each Q is heat generation due to various sources, $\vec{q}$ is the heat flux density, and ∇T is the temperature gradient. The material transport of each species i is described by

$$\nabla \cdot \vec{J}_i + \vec{u} \cdot \nabla c_i = R_i \quad (5)$$

and Fick's first law of diffusion

$$\vec{J}_i = -D_i \nabla c_i \quad (6)$$

where for each species i , J_i is its molar flux, ∇c_i is its concentration gradient, R_i is its generation, and D_i is its diffusion coefficient. Boundary conditions emerge from providing the velocity, temperature, and concentration at the entrance of the reactor. Since the COMSOL models approximate each transport phenomenon as first-order partial differential equations with no time dependence, only boundary conditions at each of the three spacial dimensions need to be specified in order for the computation to be successful.

Our model of a LPCVD reactor can be divided into three regions: the inlet characterized by

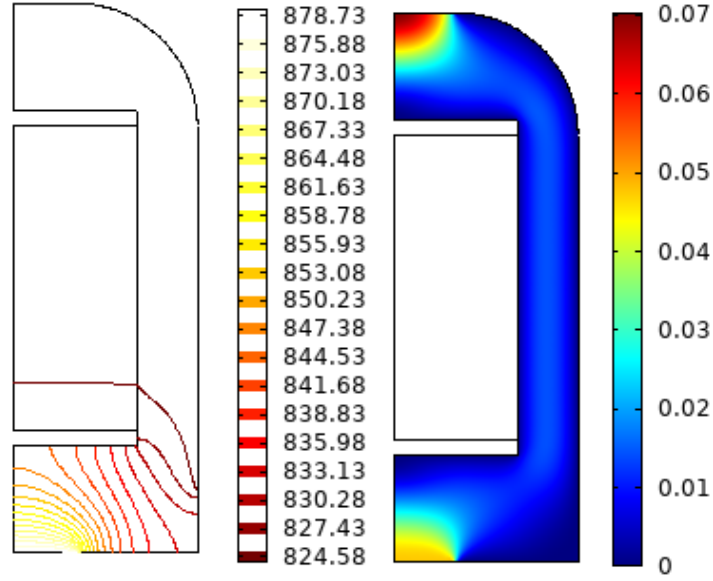


Figure 1: Temperature and velocity gradient of the reactor for an initial velocity of 4.75 cm/s, initial temperature of 880K, wafer spacing of 4.56mm, and wafer radius of 10.5 cm. Gas enters the reactor at a high temperature and enters a low-temperature zone where the vapor diffuses between the wafers.

a high temperature and high vapor flow rate, the second region occupied by the wafers with low temperature and vapor diffusion towards the center axis of the reactor, and the third region characterized by low temperature and high vapor flow rate as shown by Figure 1. In our simulation, it was assumed that convection in neither the radial nor the axial direction occur in the region of the occupied by the wafers and that the surface of this volume acts as a solid boundary for the purpose of modeling fluid convection. Heat and mass transport were simulated throughout this area as well as throughout the entire reactor, though the convective mass transport makes diffusion in the first and third regions negligible. The silane vapor enters the reactor at a high temperature and flows past the wafers at a velocity lower than at the inlet. The vapor diffuses between the wafers, and reactions occur on the wafer surfaces, forming a solid layer of silicon. Rather than using a geometric model to represent reactions on each surface of the wafers; wafer thickness, wafer diameter, and the number of wafers assuming equal spacing were defined as variables in the simulation. The silicon deposition rate is defined as the surface reaction rate divided by the molar density of silicon.

4 Methods

Using COMSOL 6.1 simulation software, an isothermal boat reactor can be used to model a LPCVD reactor. A total of five different parameters were selected based on existing LPCVD optimization research and varied in between a low and high value to measure their effect on deposition rate. The selected parameters include inlet mole fraction of SiH_4 , the temperature of section one of the reactor, the distance between wafers, the inlet velocity, and the total pressure. The LPCVD simulation is ran in accordance to the given COMSOL tutorial.⁶ From data exported off of COMSOL, a given MATLAB function is used to calculate the average deposition rate, which is used to calculate effectiveness factor. A parametric sweep was done between the low and high value for each parameter, which was $\pm 5\%$ from the optimal condition. A 2^k factorial design experiment is used to identify the most crucial factors in the effectiveness factor. The number of factors is equal to k and the resulting number of 2^k is the number of necessary treatments. A design matrix is developed to define all the treatment combinations and the dot product of each combination is taken to form the full table, or contrast constant table. Each contrast can be found using the summed value of each treatment and contrast constant table with the *sumproduct()* function in Excel. This value will be used to determine effect using the following equation,

$$Effect = \frac{Contrast}{n * 2^{k-1}} \quad (7)$$

where n is the number of replicate and k is the number of factors.

5 Results and Discussion

By variation of five different parameters, inlet mole fraction of SiH_4 (A), the temperature of section one of the reactor (B), the distance between wafers (C), the inlet velocity (D) and the total pressure (E). Each parameter was varied from a high and low setting and each combi-

nation of variables was tested at the same time to get a linear relationship for each parameter and interaction of parameters for all combinations. This was done to find the parameters that had a significant effect on the deposition rate on each wafer and the effectiveness factor. The effectiveness factor for the base model, where all parameters are at their low setting, is shown in Fig. 2. This figure shows a visualization of how data was collected radially on each wafer. The data points on each wafer were averaged to find the deposition rate per wafer, then all wafer rates were averaged for each combination of parameters, which were used for the data analysis. A 2^k factorial design of experiment analysis was done to find p-values for each parameter and second order combinations of parameters only. The parameter sets of third order and above were discarded and accounted for in the error of the experiment. This had to be done since COMSOL runs the LPCVD simulation based on governing physics equations and would give the same results for each trial. The p-values are listed in Tab. 1 and are based on the null hypotheses,

$$H_0 = \frac{VariationFactor}{VariationError} \leq 1; H_1 = Variation > RandomError \quad (8)$$

with the cutoff for a significant p-value less than .05. Which in turn means that for any p-value less than .05 the hypothesis is false and the parameter is significant and its significance can not be ruled out by random error.

5.1 Each Parameter and their Importance

5.1.1 Mole Fraction of SiH_4

The first important parameter, the mole fraction of SiH_4 in the feed, effects the rate of deposition and the effectiveness factor. This can be explained by the reactions happening in the reactor. The by increasing the amount of SiH_4 initially flowing inside the reactor increases the amount free Si atoms that are able to be deposited.⁷

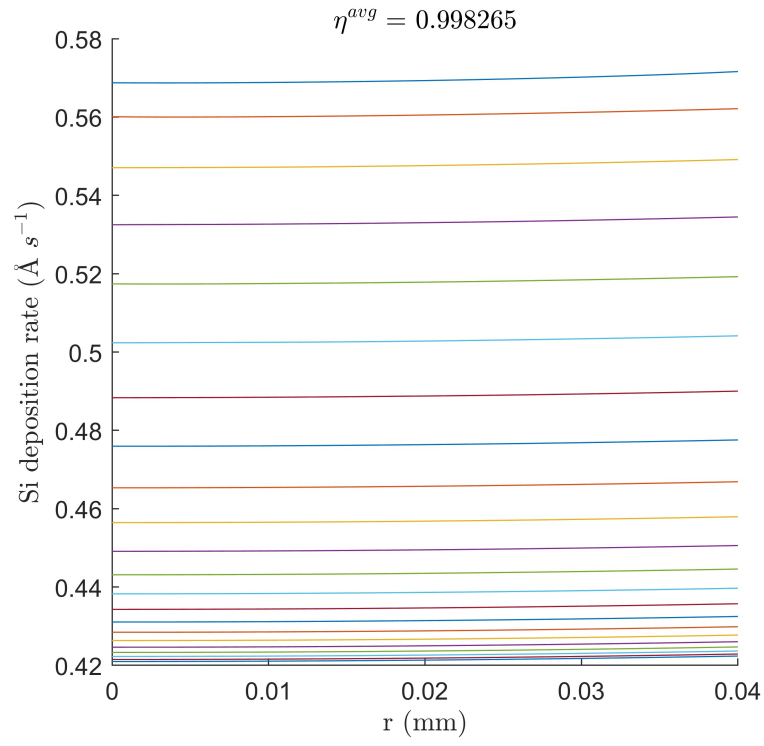


Figure 2: The deposition rate on each wafer with average effectiveness factor, where each line corresponds to a different wafer

Variable	Deposition Rate	Effectiveness Factor
	P Value	P Value
A (Molar fraction of SiH ₄)	1.876E-31	9.303E-23
B (Temperature of section 1)	1.115E-52	3.891E-34
C (Distance between wafers)	2.219E-22	3.347E-22
D (Inlet velocity)	.3821	.8475
E (Total pressure)	1.411E-28	1.118E-9
AB	1.659E-25	1.696E-17
AC	9.625E-6	1.254E-6
AD	.9640	.9825
AE	.0002248	.9519
BC	5.063E-19	1.849E-17
BD	.5952	.9092
BE	3.901E-21	.005769
CD	.9633	.9793
CE	.7542	.7280
DE	.9660	.9904

Table 1: P-values for deposition rate and effectiveness factor, where rows with double letters are the interaction of the corresponding parameters

5.1.2 Temperature and Pressure

The next two parameter with a significant effect on the system is the temperature and pressure. It has been shown by literature that as temperature or pressure increases the deposition rate also increases.⁷ Furthermore, increasing both parameters at the same time, exponentially increases the deposition rate. Our data proves that the temperature and pressure also have an effect on the effectiveness factor. This data shows that the temperature had a high interaction with all other parameters except for the inlet velocity, for effect of the deposition rate. This follows physical properties since temperature and pressure have an effect on reaction rate and each other. It should be noted that the temperature that was constrained was only the temperature of section one of the reactor. Meaning that the temperature of other sections of the reactor could vary, but even so the temperature had a significant affect. Further trials where all three chambers and the inlet stream and are constrained to one temperature should be made to see the magnitude of the affect of the temperature.

5.1.3 Wafer Spacing

The wafer spacing was another significant parameter. The interaction between wafer spacing and temperature and mole fraction were also significant given the high importance of both of those parameters. Other combinations that included that wafer spacing were not significant meaning that wafer spacing alone does not have high contribution to the deposition rate or the uniformity. This confirms logic since having many wafers closely together could hinder the uniformity and having to large a gap reduces the amount of wafers that fit inside the reactor which is less optimal. The wafer spacing does not hinder the reaction rate or the amount of free silicone hence it is not as important.

5.1.4 Inlet Velocity

The last parameter, the inlet velocity, has no effect on the system at all. Since the COMSOL simulation uses the assumption that assumes incompressible flow this could have skewed the results based on the velocity. Since the inlet flow is of gas, which is compressible, and incompressible flow should not have been assumed and a different equation should have been used.

5.2 Error

Since the data was collected using COMSOL, a simulation program, there is no benefit of running the simulation more than once. Since COMSOL outputs data governed by physics equations for each data point, there is only one corresponding value, not matter how many trials are done. Due to this, there is no random error as there would be for a real life LPCVD. Some of the random error in a live reactor could be explained by the error in the measurements of each piece of equipment. Any temperature or pressure sensors have measurement error. Also the distance between the wafers would have measurement error. Also any discrepancies in the uniformity of the wafers size and shape could cause error.

5.3 Limitations

The 2^K factorial method of analysis only is able to verify whether each parameter has an effect on the system, but can not identify trends. Some variable have already been studied to see if the deposition rate increases with their variation, but now we know what variables to change for optimization. Another limitation is that in this study the parameters only varied by five percent in either direction no trends for extremes can be made.

6 Conclusions

The parameter that had most notable impact on silicon deposition rate was inlet silane mole fraction, followed by temperature and pressure. As mentioned in 5.1.1, increasing the mole fraction of silane of the feed increases the number of free silicon atoms in the reactor volume to potentially be deposited. Upon experimenting with factors that impacted deposition rate, it would be beneficial to investigate further into the deposition uniformity. Since COMSOL simulation was used, errors can be attributed to incorrectly inputted data. The use of COMSOL also makes the replication this experiment challenging. If possible, the replication of the same LPCVD experiment should be done in a real reactor.

Bibliography

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Appendix

1.
 - Authour(s): T Setalvad et al.
 - Title: Optimization of a Low-Pressure Chemical Vapor Deposition Reactor for the Deposition of Thin Films
 - Year published: 1989
 - Journal Name: Ind. Eng. Chem. Res.

(a) In this study optimization strategies were used to improve the deposition rate of an LPCVD based on the placement of the wafers inside the reactor. Along with creating a scheme for reaction injection placement along the reactor and finding optimum temperature.
2.
 - Athours: A. Voutsas and M. Hatalis
 - Title: Structure of As-Deposited LPCVD Silicon Films at Low Deposition Temperatures and Pressures
 - Journal Name: J. Electrochem. Soc.

(a) In this study the parameters temperature, pressure, source gas dilution and deposition rate were varied to see the effect of the structure of the deposited films. It was shown that temperature alone does not control the stucture, but also the pressure and the source gas dilution.